Peatland rewetting in Schleswig-Holstein

Where are the peat soils, how are they used, how much do they emit – and where would rewetting make the most sense?

Project report · status 23 October 2026 · Bonnie Winter (GIS-Akademie Hamburg)

QGIS / open-source portfolio project · test area: Treene–Sorge lowland (≈ 304 km²)

The project in one paragraph. Drained peat soils under farmland are among the largest greenhouse-gas sources in Schleswig-Holstein. The project maps them for a test area, estimates their emissions from national emission factors, tests how robust that estimate is, derives hydrological sub-catchments from a 1 m terrain model, and then uses integer optimisation to find where a CO2 saving target can be reached with the least collateral damage. The central result: if rewetting is planned field by field, it looks efficient but would in reality wet thousands of hectares of farmland on mineral soil; planned by whole sub-catchments, the same target is reached with almost none.

Key numbers

	Value	Comment
Test area	30,358 ha	hand-drawn polygon, 34 municipalities
Organic soils	15,952 ha (53 %)	Thünen map (Wittnebel et al. 2023)
of it farmed	11,102 ha	97 % permanent grassland, 278 ha cropland
Current emissions	≈ 354,000 t CO2-eq/yr	ESTIMATE: area × national factor
Saving if all were rewetted	≈ 293,000 t CO2-eq/yr	rewetted peat still emits 5.5 t/ha
Uncertainty (90 % range)	351,000 – 358,000 t … 144,000 – 487,000 t	fields deviating independently … the whole region deviating together
Saving target used	75,000 t CO2-eq/yr	≈ ¼ of the potential
Mineral-soil farmland wetted	4,296 ha (best parcel plan) vs 52 ha (catchment plan)	same target, same peat area

Status

Step	Content	Status
00–04	inputs, overlay, emission estimate	done
01, 05	DGM1 download (355 tiles), hydrology with GRASS	done
07	optimisation: parcels vs. catchments	done
08	Monte Carlo uncertainty	done
09	atlas: 24 A4 pages, one per municipality	done
06	historical peat extent (Preußische Landesaufnahme)	next: digitising
–	write-up, GitHub repository	open

Contents

1	Question, context and study area
2	How the project is built (repository, configuration, software)
3	Data and licences
4	The analysis, step by step (00 → 09)
5	Main findings
6	Limitations of the whole approach
7	Problems we ran into and how they were solved
8	Next steps
9	Glossary
10	References

Every analysis step in chapter 4 is described in the same five parts: **why** it is needed, **what the script does**, the **results**, the **decisions** taken and why, and its **limitations**. All emission figures are estimates and are labelled as such.

1 Question, context and study area

The problem

Peat forms where water-logged conditions stop plant material from decomposing. Drained for farming, the peat is exposed to oxygen and slowly 'burns' away: the stored carbon is released as CO2, the ground subsides, and nitrous oxide is emitted as well. Raising the water table again (**rewetting**) stops most of this – but it also ends conventional farming on the land affected.

For Schleswig-Holstein, the Land's Moor-Klimaschutz information (January 2024, as summarised in the project brief) gives about 152,800 ha of carbon-rich soils (≈ 10 % of the state), of which about 129,800 ha peat (≈ 103,200 ha fen, 26,600 ha bog) and 23,000 ha Anmoor; more than 81 % are used for agriculture, and these soils cause about 17 % of the state's greenhouse-gas emissions.

The questions

- Where are the peat soils, and how are they used?
- How much do they emit – and how certain is that number?
- Which areas form their own hydrological unit, so that raising the water there does not flood neighbouring land?
- Where can a given CO2 saving be reached with the least land, the fewest units – or the least collateral damage?
- How much peatland has already been lost since the 1870s? (step 06, open)

Study area (test mode)

The analysis runs on a **test area** first, as in the earlier NRW project: a hand-drawn polygon of 30,358 ha (≈ 304 km²) in the Treene–Sorge lowland between Friedrichstadt, Hollingstedt and the A7, covering parts of the districts Schleswig-Flensburg, Nordfriesland and Rendsburg-Eckernförde (34 municipalities). It was chosen because the Thünen map shows one of the densest concentrations of fen and bog soils in the state there. Setting `TEST_MODE = False` in the configuration would run everything for the whole state.

> **Key point.** The test area is a drawn polygon, not a real river catchment. That is fine for developing and checking the method, but it cuts through natural drainage units at its edges – step 05 flags those cut-off sub-catchments.

2 How the project is built

Repository structure

```
peatland-rewetting-sh/
├─ config.py            all paths, field names, parameters, emission factors
├─ peatsh/common.py     shared helpers: AOI, clipping, GeoPackage I/O, reports
├─ scripts/             00 … 09, one numbered script per analysis step
├─ qgis/                QGIS project peatland_sh.qgz, styles, models
├─ data/raw/            downloads, never edited          (not in git)
├─ data/interim/        DGM1 GeoTIFFs, hydrology rasters  (not in git)
├─ data/processed/      peatland_sh.gpkg — all results     (not in git)
├─ outputs/             CSV tables, figures, atlas maps
└─ docs/reports/        one short markdown report per step
```

Three design rules

- **One configuration file.** Nothing is hard-coded in the scripts. File paths, field names, emission factors, thresholds and the optimisation target live in `config.py`. Changing the study area or a factor means changing one line.
- **One GeoPackage for all vector results.** Every script reads and writes layers in `data/processed/peatland_sh.gpkg`, named `<step>_<area>` (e.g. `emissions_test_catchment`).
- **A report after every step.** Each script writes a markdown report with its numbers, the parameters used and the limitations – so every result can be traced back. Scripts refuse to run with missing inputs (e.g. no cited emission factor) instead of using made-up values.

Software

Tool	Used for	Note
QGIS 3.4x LTR (OSGeo4W Shell)	Python environment for all scripts	GeoPandas 1.1.3, GDAL, SciPy 1.17 already included
GeoPandas / pyogrio	reading, reprojecting, clipping, overlay	steps 00–04, 06–08
GDAL (osgeo)	XYZ → GeoTIFF, mosaic, resampling, vectorising	steps 01, 05
GRASS r.watershed via qgis_process	flow direction, accumulation, sub-catchments	step 05
SciPy milp (HiGHS solver)	binary integer optimisation	step 07
NumPy / SciPy / Matplotlib	Monte Carlo, distribution fitting, figure	step 08
PyQGIS	layer styling, print layout, atlas export	step 09, runs inside QGIS

Why not conda / WhiteboxTools / PuLP as originally planned? Windows *Smart App Control* on this computer blocks unsigned programs. It blocked the Miniforge (conda) installer and would also block the WhiteboxTools and CBC solver executables. QGIS and its bundled Python, GDAL and GRASS are signed, so the whole pipeline was moved onto QGIS's own Python. PuLP was dropped additionally because version 4.0 changed its interface. Smart App Control was deliberately left on: switching it off cannot be undone without reinstalling Windows.

How to run

```
OSGeo4W Shell:
cd /d D:\PORTFOLIO\Peatlands\peatland-rewetting-sh\peatland-rewetting-sh
python scripts\00_check_inputs.py      (… then 01, 02, 03, 04, 05, 07, 08)
QGIS Python console (project qgis\peatland_sh.qgz open):
exec(open(r"…\scripts\09_export_maps_pyqgis.py", encoding="utf-8").read())
```

3 Data and licences

Dataset	Provider · licence	Used for	Notes
Aktualisierte Kulisse organischer Böden (Wittnebel, Frank, Tiemeyer 2023)	Thünen-Institut · CC BY 4.0	where the peat is; soil type	303,672 polygons, Germany; KAT_KURZ = soil category
Feldblöcke SH, reference date 1.1.2026	Land SH · dl-de/zero-2.0 (to confirm for 2026)	farmed land, arable vs. grassland	194,503 blocks; field HBN; CRS EPSG:4647
DGM1 (1 m terrain model)	LVermGeo SH, OpenGBD · CC BY 4.0	hydrology	XYZ text tiles 1×1 km; bulk-download index of 18,685 tiles
VG250 boundaries (state, districts, municipalities)	BKG · dl-de/by-2.0	statistics per municipality, atlas	fields GEN, AGS, SN_L, GF
Emission factors	Tiemeyer et al. 2020, Ecol. Indic. 109:105838	emission estimate, Monte Carlo	Table 5 (totals), Table 2 (ranges)
OpenStreetMap	© OSM contributors · ODbL	background maps	greyed in the atlas
Preußische Landesaufnahme (Chronologen)	LVermGeo SH · CC BY 4.0 (to check)	historical peat extent (step 06)	service address still to be found
Protected areas (FFH, NSG)	LfU SH · to check	not used yet	–

Coordinate reference system

Everything is processed in **EPSG:25832** (ETRS89 / UTM zone 32N), the system of the LVermGeo SH data. The Feldblöcke arrive in **EPSG:4647**: the same projection, but with the zone number '32' written in front of the easting (32 500 000 instead of 500 000). The scripts reproject automatically.

Attribution required on every map

© GeoBasis-DE/LVermGeo SH/CC BY 4.0 (DGM1) · © GeoBasis-DE/BKG (VG250) · Thünen-Institut (organic soils) · Land Schleswig-Holstein (Feldblöcke) · © OpenStreetMap contributors. The atlas prints these in its footer.

4 The analysis, step by step

The order below follows the logic of the analysis, not the script numbers: data → overlay → emissions → uncertainty → hydrology → optimisation → maps.

Step 00 – Check inputs

Why

Before any analysis it must be clear what is actually in each file: its coordinate system, its attribute fields and their values. Field names differ between datasets and years, so they are looked up, not assumed.

What the script does

- lists every input in `config.INPUTS`: exists?, number of features, CRS, geometry type
- shows every attribute field with five example values
- reports how many DGM1 GeoTIFFs are present

Results (test area)

All main inputs were found. The report gave the field mapping now in the configuration: soil type **KAT_KURZ** (NH fen, HH bog, MF former peat soil, DK_* peat under a mineral cover), land use **HBN** (Ackerland / Dauergrünland), parcel ID **FLIK**.

Decisions and why

- Count features from the file header (pyogrio.read_info) instead of loading 300,000 polygons.
- Do not use the Feldblöcke field FLAECHE: it is text with a decimal comma. Areas are recomputed from geometry everywhere.

Limitations

- Shows only a sample of 500 rows for example values – rare codes can be missed (step 03 therefore lists every land-use value it could not map).

Step 02 – Prepare inputs

Why

All layers must share one coordinate system and be cut to the study area before they can be overlaid.

What the script does

- loads the AOI polygon and writes it to the GeoPackage
- reads each input **only inside the AOI** (spatial mask – much faster for Germany-wide files)
- repairs invalid geometries, reprojects to EPSG:25832, clips, recomputes areas

Results (test area)

Layer	Features	Area (ha)
AOI	1	30,358
organic soils	1,470	15,952
Feldblöcke	7,315	21,922
municipalities	34	30,358

Decisions and why

- Thünen's own AREA_HA field is dropped and recomputed: it is wrong after clipping, and GeoPackage treats area_ha and AREA_HA as the same field name.
- keep_geom_type when clipping, so stray lines/points from the cut don't end up in polygon layers.

Limitations

- Protected areas are not yet included.

Step 03 – Organic soils × agricultural land use

Why

Emissions depend on how the peat is used. The overlay answers: how much peat lies under arable land, how much under grassland, and in which municipality?

What the script does

- maps HBN codes to classes: Ackerland → cropland, Dauergrünland → grassland
- intersects Feldblöcke with organic soils, then with municipalities
- summarises hectares by soil type × land use and per municipality

Results (test area)

11,102 ha of the 15,952 ha organic soils (70 %) are farmed – **97 % of it permanent grassland** (10,824 ha), only 278 ha cropland. 104 'non-agricultural eligible areas' and one permanent crop were excluded.

Soil class	Cropland ha	Grassland ha
NH – fen	79	4,197
MF – former peat soil	112	1,773
HH – bog	3	1,693
DK_FL_NH – fen, shallow mineral cover	19	1,684
DK_MA_NH – fen, thick mineral cover	57	1,085
DK_FL_HH – bog, shallow mineral cover	8	392

Decisions and why

- Land use from the Feldblöcke (agricultural reference parcels), not from ATKIS: they are the official, current (2026) record of what is farmed.
- Very small overlay slivers are dropped.

Limitations

- A Feldblock is not an ownership parcel – one block can belong to several owners or farmers.
- The municipality top-10 table in the current report 03 was still sorted by cropland only (bug fixed; re-run step 03 to refresh the report).

Step 04 – Emission estimate

Why

The core quantity of the project: how much greenhouse gas do the farmed peat soils emit, and how much would rewetting save?

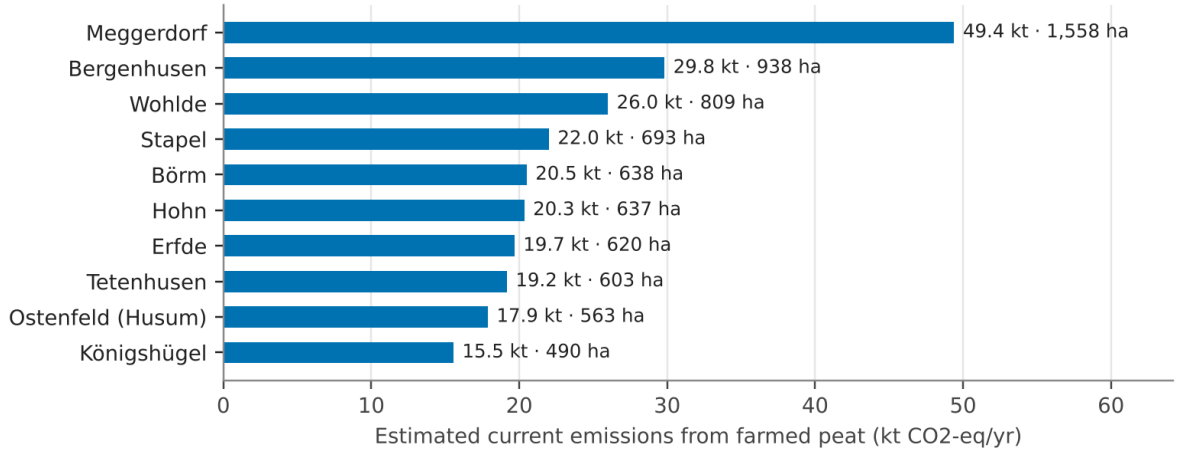
What the script does

- assigns each overlay piece the emission factor of its land-use class
- emissions = area × factor; saving = area × (factor − factor for rewetted peat)
- sums per municipality

> **Formula.** $E = \Sigma_c A_c \times EF_c \cdot \text{saving} = \Sigma_c A_c \times (EF_c - EF_{\text{rewetted}})$ with A = farmed peat area (ha), EF = emission factor, c = cropland / grassland.

Results (test area)

≈ 354,000 t CO2-eq/yr current emissions; **≈ 293,000 t** could be saved if all farmed peat were rewetted (≈ 61,000 t remain, mainly methane from wet peat). Meggerdorf alone accounts for ≈ 49,000 t.



Estimated current emissions by municipality (top 10). Because almost everything is grassland with one factor, the ranking simply follows the farmed peat area.

Class	Factor (t CO2-eq/ha/yr)	Source
cropland on organic soil	40.4	Tiemeyer et al. 2020, Table 5
grassland on organic soil	31.7	Tiemeyer et al. 2020, Table 5
rewetted organic soil	5.5	Tiemeyer et al. 2020, Table 5

Decisions and why

- **Implied national emission factors** from the German greenhouse-gas inventory method: they include CO2 (incl. dissolved organic carbon), methane from soil and ditches and nitrous oxide, converted with the IPCC AR4 warming potentials, and they already contain the undrained share of each land use – so they can be multiplied with area directly.
- The same factor is applied to all soil classes incl. former peat soils and covered peat: the paper found no emission difference between 'true peat' and lower-carbon organic soils at German farm sites (sect. 4.1.1).
- The script refuses to run without cited factors.

Limitations

- National averages: the actual water table, peat thickness and management of each field are ignored.
- Parts of the test area are in wet-grassland schemes; there the estimate is probably too high.

Step 08 – Monte Carlo: how certain is the estimate?

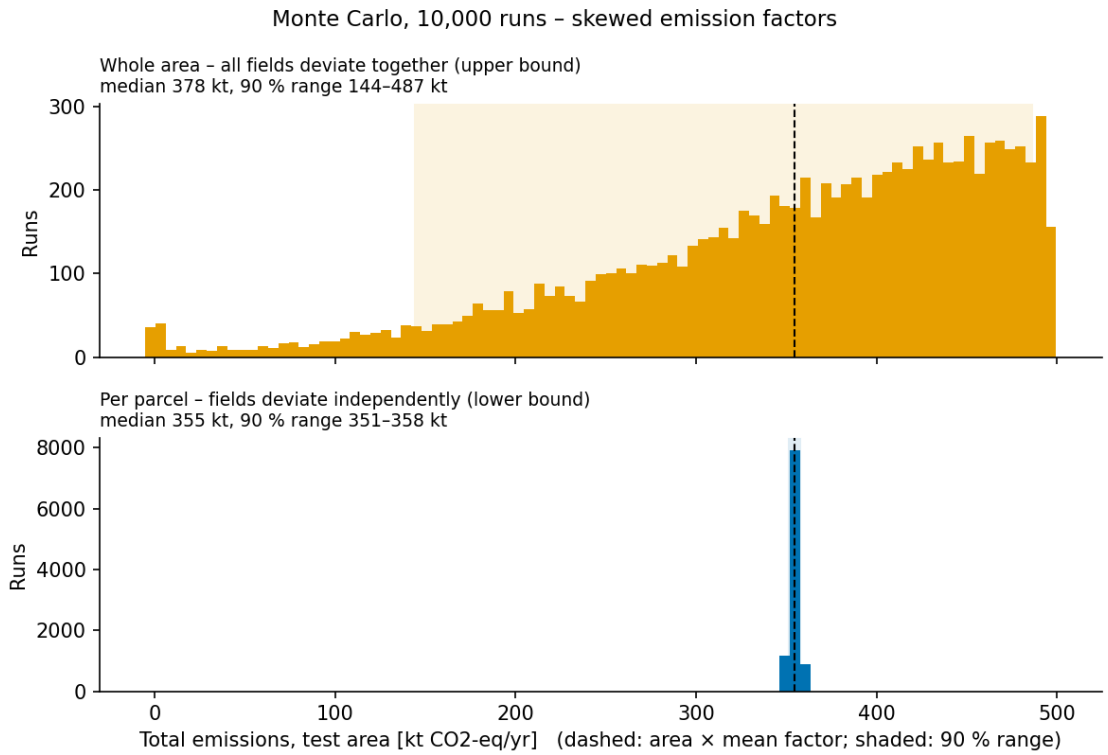
Why

A single number like 354,000 t hides its uncertainty. The question is how wide the plausible range is – and what drives it.

What the script does

- draws emission factors 10,000 times from a distribution fitted to the paper's numbers
- recomputes total emissions and savings for every draw
- runs two scenarios: **whole area** (one draw for all fields) and **per parcel** (an independent draw for each of 4,438 parcel/land-use units)

Results (test area)



Monte Carlo, 10,000 runs. Top: all fields deviate together. Bottom: fields deviate independently. Same x-axis, so the widths compare directly.

Scenario	Emissions median	90 % range	Savings median	90 % range
whole area (upper bound)	377,750	143,853 – 486,858	313,018	76,205 – 447,116
per parcel (lower bound)	354,681	350,943 – 358,406	293,210	289,232 – 297,117

Decisions and why

- **The key insight.** The paper's ranges (Table 2) are percentiles over all grid points: they describe how much *single hectares* differ, mainly because of their water level – not how uncertain the *average* over a large area is. A first version that applied one random factor to all fields therefore greatly overstated the uncertainty.
- **Lopsided distribution.** For drained land, most hectares emit close to a ceiling (CO2 stops rising once a field is deeply drained) with a long tail towards wetter fields. A two-piece normal distribution is fitted so that its mean equals the inventory factor exactly and its 2.5/97.5 % points match the paper within ≈ 0.7 t C/ha.
- **Physical floor.** Even a fully wet field cannot emit less than the lower limit of the paper's CO2 response curve (−0.93 t C/ha/yr); without this cap the tail produced impossible values.
- **Two bounding scenarios** instead of one guess about how strongly fields are correlated.

Limitations

- Only emission-factor variability is included – not mapping errors, the uncertainty of the response functions themselves, or the choice of warming potentials.
- Reality lies between the two scenarios: neighbouring fields share ditches and pumps.

Key point. The emission factors themselves don't make the estimate uncertain – the unknown **regional water level** does. If the whole region deviates from the national average, the range is −60 % / +35 %; if fields deviate independently, it is ±1 %. Local water-level data would narrow this down.

Step 01 – DGM1 terrain model

Why

Hydrology needs a detailed terrain model; in flat fen landscapes a few centimetres decide where water flows. The 1 m DGM1 of Schleswig-Holstein is open data.

What the script does

- works out which 1×1 km tiles touch the AOI (355)
- downloads them via the portal's official **bulk-download index** (a GeoJSON with a direct link per tile), one at a time with a 1-second pause
- converts each XYZ text tile (1,000,000 lines of x y z) straight into a compressed 1 m GeoTIFF – the 28 MB text files are never stored; ≈ 0.6 s per tile
- records each tile's survey year and writes a tile layer (present / missing / year)

Results (test area)

All **355 tiles** complete. The survey years form clear blocks: west and centre mostly **2025** (205 tiles), a northern strip **2024** (39), the east (easting ≥ 528 km) **2005/2006** (106), plus 2017 (1) and 2021 (4).

Decisions and why

- The interactive download client allows only one tile per order; the bulk index made the download automatic and reproducible – and it is the sanctioned route for mass downloads.
- HTTPS certificates are checked against the Windows certificate store (package truststore); certificate checking was never switched off.
- Safe to re-run: finished tiles are skipped; failures are listed.

Limitations

- Tiles almost 20 years apart meet at easting ≈ 528 km. Drained peat subsides by roughly 1 cm per year, so steps of centimetres to decimetres can appear at tile edges – relevant in flat terrain.

Step 05 – Hydrology with GRASS r.watershed

Why

Rewetting raises the water level in a whole hydrological unit, not in a single field. To plan realistically we need those units: sub-catchments and the streams that connect them.

What the script does

- mosaics the 355 tiles and resamples them to a 5 m working grid
- runs GRASS **r.watershed** through qgis_process: flow direction (D8), flow accumulation, streams, sub-catchments of at least 50 ha (20,000 cells)
- vectorises the sub-catchments, computes their share of organic soil and flags catchments touching the AOI/DEM edge as incomplete

Results (test area)

	Run 1 (mean)	Run 2 (min, used)
sub-catchments	393	367
incomplete (touch the edge)	80	80
complete and ≥ 50 % organic soil	207	196 (14,741 ha; 12,119 ha peat)

Visual check: the catchment boundaries run largely along dykes, dams and road embankments – realistic for a polder landscape, where these are the true water divides.

Decisions and why

- **r.watershed** routes water through depressions with a least-cost search, so no 'fill sinks' step is needed. Filling would create large artificial flats in fen terrain.
- **Minimum instead of mean** when resampling 1 m → 5 m. Ditches are only 1–3 m wide; averaging smoothed them away and streams wandered across fields in stair-steps. Taking the lowest 1 m cell keeps ditch bottoms connected – the streams now follow the ditch pattern.
- Outputs are named after the method (stream_min.tif …), so both runs can be compared in QGIS.

Limitations

- **Pumping stations.** The polders are drained by ditches and pumps that lift water into the embanked Alte Sorge. A terrain model cannot see pumps, culverts or pipe drains, so the main river is not recognised as the main drain.
- The real drainage units are the pumping-station catchments of the water and soil associations (Wasser- und Bodenverbände); the DEM catchments approximate them.
- Possible improvement: 'stream burning' (GRASS r.carve) with the mapped ditch network.

Step 07 – Optimisation: parcels vs. catchments

Why

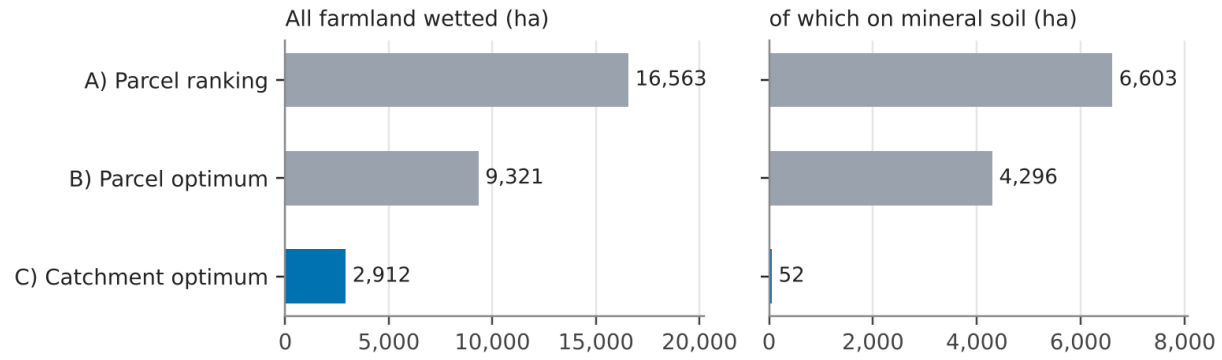
A ranking ('best fields first') is the intuitive way to choose areas. The question is whether an optimisation does better – and what changes when hydrology is taken into account.

What the script does

- **A) Parcel ranking:** Feldblöcke sorted by saving per ha, taken until the target is reached
- **B) Parcel optimum:** binary integer programme over 3,411 parcels – least area that reaches the target
- **C) Catchment optimum:** the same over 272 complete sub-catchments – a catchment is chosen whole; the objective is to wet as little farmland on **mineral** soil as possible
- for A and B it also counts what would really get wet: all farmland in every catchment that contains a chosen parcel

Integer programme. Choose $x_i \in \{0, 1\}$ for every unit i (parcel or catchment): minimise $\Sigma\ \text{cost}_i \cdot x_i$ subject to $\Sigma\ \text{saving}_i \cdot x_i \geq \text{target}$. Cost = area (B) or mineral-soil farmland in the catchment (C).

Results (test area)



All three approaches save ≈75,000 t CO2-eq/yr. Parcel approaches: farmland in every catchment that contains a chosen parcel.

Approach	Units	Peat rewetted ha	Saving t/yr	Catchments	Farmland wetted ha	on mineral soil ha
A) ranking	914 parcels	2,791	75,037	230	16,563*	6,603*
B) parcel optimum	808 parcels	2,789	75,001	119	9,321*	4,296*
C) catchment optimum	82 catchments	2,860	75,007	82	2,912	52

* if the water level is raised in every catchment containing a selected parcel – which is what rewetting physically does.

Decisions and why

- Target 75,000 t CO2-eq/yr ≈ a quarter of the potential: large enough that the solver has to choose.
- Solver: SciPy milp (HiGHS) – no external program, all three models solved to proven optimality in seconds.
- Without hydrology, A and B are almost identical (2,791 vs 2,789 ha): with only two emission factors nearly every hectare saves the same, so optimisation cannot beat a ranking – an important result in itself.
- Each parcel is assigned to the catchment containing its interior point, so parcels touching a boundary are not counted twice (this corrected A and B by ≈ 10–15 %).
- Objective switchable in config.py: non_peat_farmland (default), affected_farmland, count.

Limitations

- Uses the DEM catchments (see step 05). With pumping-station catchments the result could change.
- No costs, ownership, protected areas or farmers' willingness yet.
- Savings are estimates with the uncertainty from step 08.

Key point. Planned field by field, rewetting looks efficient but would wet 4,000–6,600 ha of farmland on mineral soil that emits nothing. Planned by sub-catchment, the same 75,000 t are reached with 52 ha – about 80 times less collateral damage. Hydrology turns the optimisation from a formality into the core of the analysis.

Step 09 – Atlas (PyQGIS)

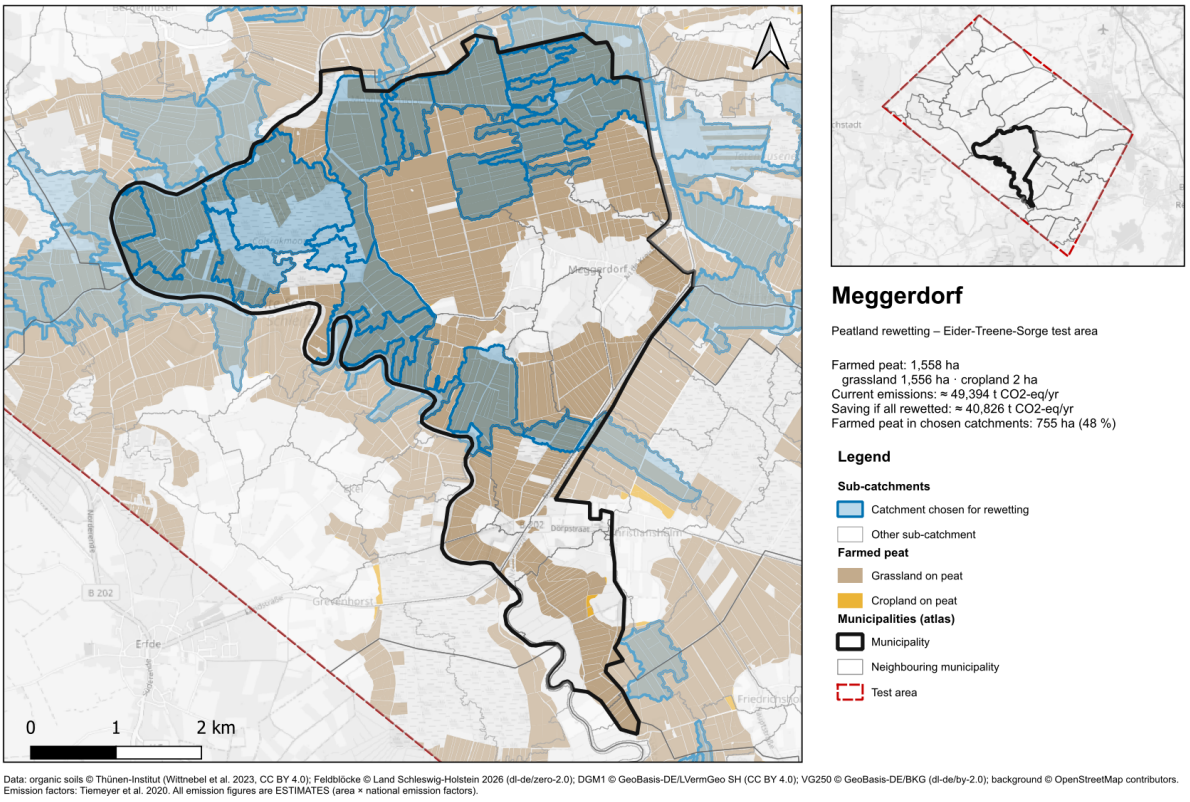
Why

Results have to be communicated per municipality – the level at which planning and talks with landowners happen.

What the script does

- builds a coverage layer: the 24 municipalities with at least 50 ha farmed peat, with key figures
- loads and styles all map layers in a separate group (your own layers stay untouched)
- creates the print layout 'Atlas_Gemeinden' (A4 landscape): main map following each municipality, overview map, title, key figures, legend, scale bar, north arrow, sources
- exports one PDF with all pages plus one PNG per municipality, biggest emitters first

Results (test area)



Atlas page for Meggerdorf. Blue: sub-catchments chosen by the catchment optimum; brown: grassland on peat; black: the municipality.

Decisions and why

- Everything scripted, so the maps can be re-made after any analysis change in one run.
- Legend fonts set explicitly (QGIS defaults are far too large for an 84 mm column); English number format; sources in a strip along the page bottom.
- zoomToExtent instead of setExtent – the latter silently resizes the map frame.

Limitations

- The PDF is still ≈ 77 MB (many detailed polygons on 24 pages); for GitHub or e-mail, a reduced version is needed.
- Re-running the script overwrites manual changes in the layout.

5 Main findings

- **Most of the peat is farmed, almost all as grassland.** 70 % of the organic soils in the test area are in Feldblöcke, 97 % of that permanent grassland. The cropland/grassland split therefore explains little; what differs between fields is mainly their water level.
- **The order of magnitude is large.** ≈ 354,000 t CO2-eq/yr from ≈ 11,100 ha – about 32 t per hectare and year, consistent with a grassland-dominated area under the national method.
- **The uncertainty is a data gap, not a statistical problem.** The estimate is stable (±1 %) if fields deviate independently, but could be −60 % / +35 % if the whole region is wetter or drier than the German average. Regional water-level data are the key to a better estimate.
- **Without hydrology, optimisation adds nothing.** With uniform factors, ranking and optimisation choose the same area; any apparent 'priority list' of fields is false precision.
- **With hydrology, the picture changes completely.** Planning by sub-catchment reaches the same target while wetting ≈ 80 times less farmland on mineral soil (52 ha vs 4,296 ha) – concentrated in 82 units that are almost pure peat.
- **Data quality matters in flat terrain.** Terrain-model resampling (minimum vs mean) and survey dates 20 years apart visibly affect streams and catchments.

Suggested sentences for the README

'In drained fen landscapes, a field-by-field selection of rewetting areas is misleading: water levels can only be raised for whole hydrological units. Optimising by sub-catchment reaches a saving of 75,000 t CO2-eq/yr while affecting only 52 ha of farmland on mineral soil, compared with about 4,300 ha for the best field-based selection.'
'The emission estimate is robust to the variability of emission factors between fields (±1 %), but not to a systematic regional deviation of the water table (−60 % / +35 %).'

6 Limitations of the whole approach

Limitation	Effect	How it could be addressed
National average emission factors	local water level, peat thickness, management ignored	water-level map (Thünen), field data, response functions
DEM-based catchments in pumped polders	pumps, culverts, pipe drains invisible	pumping-station catchments of the water boards; stream burning
Hand-drawn test area	80 sub-catchments cut at the edge	use a real catchment as AOI
Mixed DGM1 survey years	possible steps at tile edges	map and discuss; check edges
Feldblock ≠ ownership parcel	no statement about owners	ALKIS (not open)
No costs, protected areas, willingness	optimum is purely physical	add constraints / costs
Monte Carlo covers factors only	mapping errors not included	add area uncertainty

7 Problems we ran into and how they were solved

These are worth knowing – several make good interview stories about debugging and judgement.

Problem	Cause	Solution
Inputs reported as MISSING	data sat one or two folders deeper after unzipping	paths adjusted in config.py
ValueError when counting features	empty column list not allowed by pyogrio	count from file header (read_info)
'Error adding field area_ha'	GeoPackage field names are case-insensitive (AREA_HA exists)	drop old field before recomputing
Miniforge installer failed	Smart App Control (Device Guard) blocks unsigned programs	use QGIS's bundled Python
PuLP TypeError (cat=…)	PuLP 4.0 changed its API; CBC solver would be blocked anyway	SciPy milp with HiGHS
Uncertainty looked 'massive'	per-hectare ranges applied as area uncertainty	two-piece normal + two correlation scenarios
Impossible negative emissions	unbounded lower tail of the distribution	floor at CO2-Cmin = −0.93 t C/ha/yr
Only one DGM1 tile per order	deliberate limit of the download client	official bulk-download GeoJSON index
CERTIFICATE_VERIFY_FAILED	QGIS's Python doesn't use the Windows certificate store	truststore package (checking stays on)
r.watershed 'not found'	qgis_process loads plugins separately; LTR command name	qgis_process-qgis-ltr plugins enable grassprovider
Streams across fields	mean resampling erased 1–3 m ditches	minimum resampling
Parcels counted in two catchments	touching boundaries counted as intersecting	assign by interior point
Atlas inset too tall, legend overlapping	setExtent resizes the frame; default legend fonts	zoomToExtent; compact legend

8 Next steps

- **Step 06 – historical peat extent:** find the Chronologen service URL in SH-MIS (record 233ceb49-942a-40b4-8263-87845c2a1857), check the licence, digitise the moor signature of edition 1 (1877–80) for one or two sheets around Meggerdorf/Bergenhusen into data/raw/historical/peat_1880_digitised.gpkg (fields hist_type, note), then run step 06.
- Re-run step 03 so its report has the corrected municipality table.
- Optional: stream burning; pumping-station catchments; protected areas as a constraint; a Monte Carlo of the catchment result; a smaller atlas PDF.
- Write-up: README with the main finding and figures, a short report, data-quality figure (DGM1 survey years), before/after streams.
- Put the code under version control on GitHub (data and large outputs excluded).

9 Glossary

Term	Meaning
Organic soil / peat	Soil with a high share of organic matter, formed under water-logged conditions. Niedermoor = fen (fed by groundwater), Hochmoor = bog (fed by rain), Anmoor = transitional soil with less organic matter.
Feldblock (FLIK)	Agricultural reference parcel used for EU subsidies; a block of farmland bounded by permanent features. Not an ownership parcel.
CO2-equivalent, GWP	Other gases converted into the CO2 amount with the same warming effect over 100 years. IPCC AR4 values used here: CH4 = 25, N2O = 298.
Implied emission factor (IEF)	Average emission per hectare for a land-use category including all gases and its undrained share, as used in the national inventory.
DGM1 / DEM	Digital terrain model with 1 m cells: ground height without buildings and vegetation, from airborne laser scanning.
Resampling (min / mean)	Combining fine cells into coarser ones. Mean averages them; minimum keeps the lowest value, preserving narrow channels.
D8 flow direction	Each cell drains to the steepest of its 8 neighbours.
Flow accumulation	Number of upstream cells draining through each cell; high values mark streams.
Sub-catchment	Area draining to one stream segment; here at least 50 ha.
Polder, pumping station	Low-lying land protected by dykes, drained by ditches; pumps lift the water into the embanked river.
Integer programme (MILP)	Optimisation where decisions are whole numbers – here yes/no (0/1) per parcel or catchment.
HiGHS	Open-source optimisation solver built into SciPy.
Greedy ranking	Take the best-scoring units one by one until the target is reached; not guaranteed optimal.
Monte Carlo	Repeat a calculation many times with randomly drawn inputs to see the spread of results.
Two-piece normal	A distribution with different spreads below and above its peak – fits lopsided ranges.
Atlas (QGIS)	A print layout that repeats for every feature of a coverage layer – here one page per municipality.
qgis_process	Command-line tool that runs QGIS Processing algorithms (incl. GRASS) from scripts.
EPSG:25832 / 4647	ETRS89 / UTM 32N; 4647 is the same projection with the zone number '32' prefixed to the easting.

10 References

- Tiemeyer B., Freibauer A., Borraz E. A. et al. (2020): A new methodology for organic soils in national greenhouse gas inventories: Data synthesis, derivation and application. Ecological Indicators 109: 105838. https://doi.org/10.1016/j.ecolind.2019.105838
- Wittnebel M., Frank S., Tiemeyer B. (2023): Aktualisierte Kulisse organischer Böden in Deutschland. Thünen Working Paper 212. Data: https://doi.org/10.3220/DATA20230510130443-0
- Landesamt für Vermessung und Geoinformation Schleswig-Holstein: Offene Geobasisdaten (OpenGBD), DGM1; download portal https://geodaten.schleswig-holstein.de/gaialight-sh/_apps/dladownload/
- Land Schleswig-Holstein: Feldblöcke, reference date 1.1.2026; metadata https://gdk.gdi-de.org/geonetwork/srv/api/records/21f67269-780f-4f3c-8f66-03dde27acfe7
- Bundesamt für Kartographie und Geodäsie: Verwaltungsgebiete 1:250 000 (VG250), https://gdz.bkg.bund.de
- Land Schleswig-Holstein (2024): Moor-Klimaschutz – figures as summarised in the project brief.
- GRASS GIS r.watershed manual; QGIS documentation (Processing, qgis_process, print layouts / atlas).
- SciPy: scipy.optimize.milp (HiGHS solver).